higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 3

(12030263)

7 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Electrical faults occur when electrical wiring, cables and fittings are not installed correctly or maintained correctly, according to basic safety requirements.

State FIVE ways in which the risk of a fire caused by electrical faults, can be reduced. **[5]**

QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

2.1.1 Different types of knots are used when lifting heavy objects.

2.1.2 Lifting machines such as cranes and forklifts must only be operated by trained and qualified people.

2.1.3 During lifting of heavy objects, it is very important for everyone around the site to be alert and aware of the process.

2.1.4 Lifting machines such as forklifts must be fitted with a non-fail safe device.

2.1.5 Lifting equipment such as overhead gantry cranes must not be fitted with an end stop because that will slow down the effectiveness of the equipment.

(5 × 1) (5)

2.2 Name FOUR body parts that can be injured when lifting heavy objects manually and incorrectly. (4)

2.3 State SEVEN typical hand signal instructions that a banksman can give to the crane operator during a lifting process. (7)

2.4 Name FOUR types of knots that can be used in lifting. (4)

[20]

QUESTION 3

3.1 Give the definition of the term w*ork.* (2 × 1) (2)

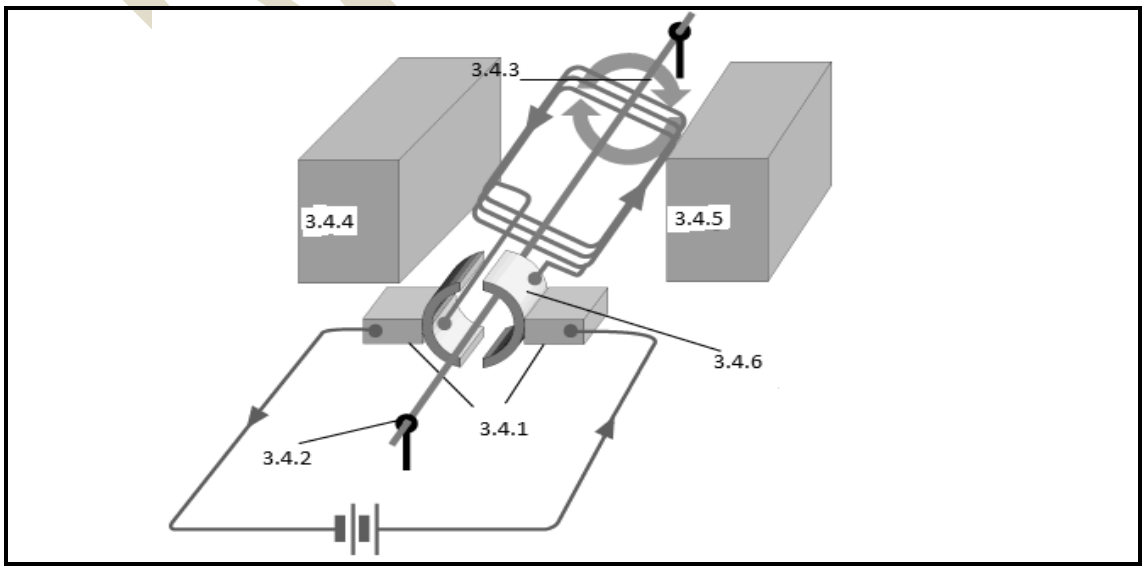
3.2 Differentiate between *thermal energy* and *kinetic energy.* (2 × 2) (4)

3.3 Choose a description from COLUMN B that matches a/an word/item in COLUMN A. Write only the letter (A–H) next to the question number (3.3.1–3.3.8) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.3.1	Torque	A	rate of doing work
3.3.2	Head gasket	B	provide a good seal
3.3.3	Work	C	force acting over a distance
3.3.4	Connecting rod	D	connects the piston to the crankshaft
3.3.5	Crankcase	E	lower part of the engine
3.3.6	Carburettor	F	a devise on petrol engines to make a mixture of air with petrol
3.3.7	Lubricating oil	G	reduces friction
3.3.8	Power	H	twisting force

(8 × 1) (8)

3.4 The sketch below shows a diagram of a basic electric motor. Name each part/component indicated by writing only the answer next to the question number (3.4.1–3.4.6) in the ANSWER BOOK.



(6)
[20]

QUESTION 4: TRANSMISSIONS

4.1 Briefly explain the process of transmissions in machinery. (3 × 1) (3)

4.2 Choose the parts/components of a clutch from the list of parts given below. Write only the answer next to the question number (4.2) in the ANSWER BOOK.

cover plate; releasing plate; pressure plate; diagnostic plate; release plate; diaphragm spring; clutch plate; synchronising plate; reverse shaft plate; idling plate

(5 × 1) (5)

4.3 What is cheaper to replace, the gearbox or the brake pads? (1)

4.4 Name ONE application of a diesel electric drive transmission. (1)

[10]

QUESTION 5: MAINTENANCE OF VEHICLES

5.1 It is important that construction vehicles are kept in a good condition at all times.

Briefly explain the procedure to be followed when changing the engine oil in a vehicle. (7 × 1) (7)

5.2 The following sentences describe the maximum projection of loads and the testing of vehicle batteries using a multimeter. Choose the correct option from those given in brackets and write the answer next to the question number (5.2.1–5.2.5) in the ANSWER BOOK.

The regulation on projections states that any projection may not be more than 5.2.1 (300 mm/450 mm) beyond the front of the vehicle and not more than (5.2.2 … 1,8 m/2,5 m) beyond the back of the vehicle.

Some types of multimeters have a display light and LED light which scan a battery's voltage and show a 5.2.3 (blue light/green light) to indicate that the battery is fully charged.

The 5.2.4 (red light/yellow light) indicates that the battery is very low while the 5.2.5 (red light/green light) shows that the battery is flat. (5 × 1) (5)

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5.3 Regular maintenance checks and services are required to make sure that construction vehicles operate properly and safely.

Briefly indicate how often the following services should be carried out on vehicles. Write only the answer next to the question number (5.3.1–5.3.6) in the ANSWER BOOK.

ACTIVITY	CHECKS AND SERVICES INTERVALS
Example: Tyre pressure	**Check weekly**
Fuel levels	**5.3.1 …**
Oil levels	**5.3.2 …**
Coolant levels	**5.3.3 …**
Brake fluid levels	**5.3.4 …**
Battery water levels	**5.3.5 …**
Changing engine oil, air and fuel filters	**5.3.6 …**

(6 × 1) (6)

5.4 Name TWO tools that will be needed when changing a vehicle's wheel. (2)
[20]

QUESTION 6: SMALL PLANT AND POWER TOOL OPERATION

6.1 Briefly explain why portable generators are used. (2 × 1) (2)

6.2 Discuss FOUR possible problems when overloading a concrete mixer. (4)

6.3 State SIX safety guidelines to follow when working with an electric hand tool. (6)

6.4 Name THREE parts/components of an angle grinder. (3)
[15]

QUESTION 7: OXYACETYLENE

7.1 Define the term *regulator.* (2 × 1) (2)

7.2 Name FOUR components of oxyacetylene equipment. (4)

7.3 Briefly explain the correct storing and lashing of acetylene cylinders. (4 × 1) (4)
[10]

TOTAL: 100

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